

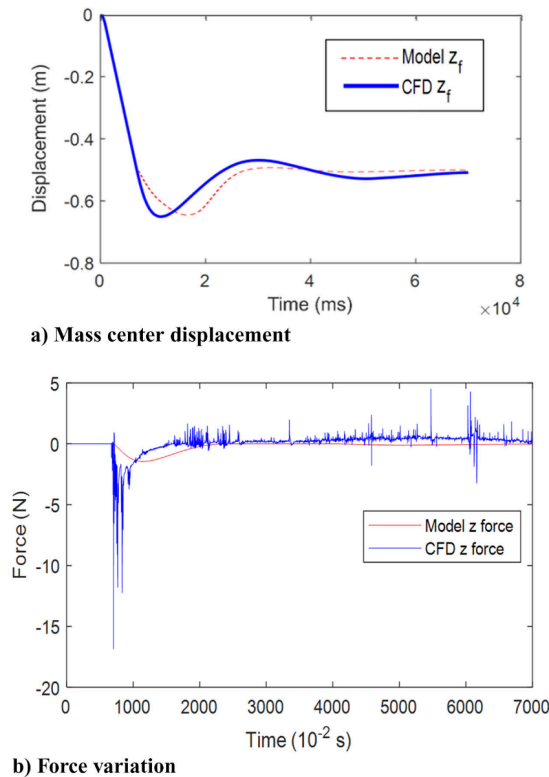


Fig. 17 Model and CFD results comparison.

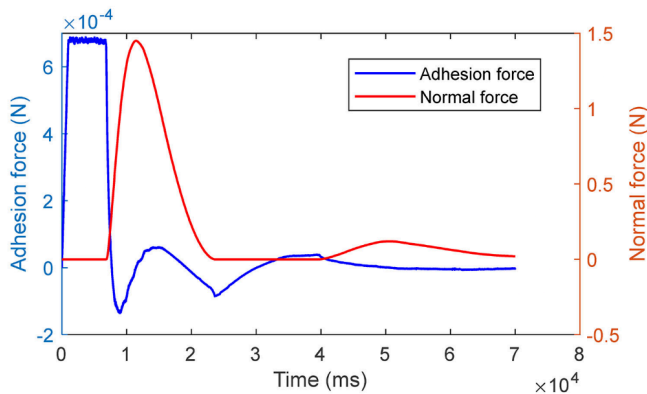


Fig. 18 Collision's normal force and adhesion force.

VII. Conclusions

This research is introducing an innovative mechanical model for simulating the behavior of liquid propellant, specifically, addressing the witnessed phenomenon of the wobbling spherical bulks of propellant in zero gravity. By exploiting the dominance of the surface tension to create an elastic body analogy and employing an enhanced Hertz contact theory, the complex fluid dynamics is simplified into more manageable mechanical principles. The proposed approach, using the TAB method and Hook's law, allows the elastic model to incorporate the real properties of the liquid, enhancing its accuracy and reliability for future studies which is particularly important in capturing the dynamic interactions of the fuel wobbling masses.

This simplified mechanical model overcomes the geometrical and mechanical restrictions imposed by the traditional models; it allows us to analyze the motion of the fuel with all its six DOF, offering greater flexibility and precision in exhibiting the real dynamics of the propellant. Results of simulations successfully imitated the on-space witnessed phenomena of the wobbling liquids interacting with both hydrophobic and adhesive walls. Demonstrated, as well, is the

effect of the rigid-flexible coupled dynamics on an orbiting spacecraft underscoring the imperative need for such manageable mechanical model allowing the design of robust controllers. In addition to qualitatively presenting the dynamics of the surface-tension-dominated liquid bulks, the results of simulations presented reasonable forces and displacements magnitudes in big agreement with the CFD results. Traditional models have often treated fuel disturbances as continuous, assuming constant contact with tank walls and a consistent system mass. However, the proposed model addresses the unpredictable and intermittent disturbances, as evidenced by collision forces. The disturbances can be continuous when the propellant is in continuous contact with the tank wall, and discontinuous when the propellant is bouncing between walls. The second phenomenon has been observed in space but not previously mechanically modeled until the introduction of this model. The enhanced Hertz contact model enables the presentation of these dynamics. Future work will focus on refining the model, adjusting its parameters and exploring its application in various spacecraft configurations, paving the way for new advancements attitude controllers design for challenging futuristic projects such as space refueling stations and liquids loading in general.

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